

Adeesha Perera - Technical Skills

Machine Learning Engineer | Data Scientist

Machine Learning & Data Science

Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, CatBoost, Hugging Face Transformers

MLOps: MLflow, DAGsHub, Apache Airflow, Feast Feature Store, GitHub Actions CI/CD

Used in: Real-Time Fraud Detection System

Techniques: Supervised Learning, Feature Engineering, Model Optimization, A/B Testing

Evaluation: ROC-AUC, PR-AUC, F1-Score, Cross-Validation, RAGAS (RAG Evaluation)

Used in: Hybrid RAG Portfolio Assistant

Streaming: Apache Kafka, Real-time Feature Computation

Used in: Real-Time Fraud Detection System

Monitoring: Grafana, Prometheus, Evidently AI (Drift Detection)

Used in: Real-Time Fraud Detection System

NLP & Large Language Models

LLMs: GPT-OSS-120B, Llama 3.3, Groq Inference, Gemini Embeddings

Used in: Hybrid RAG Portfolio Assistant

RAG: Hybrid Search (BM25 + k-NN), Cohere Reranking, Parent-Child Chunking

Used in: Hybrid RAG Portfolio Assistant

Document Extraction: Docling (PDF/DOCX/PPTX/HTML), Layout Analysis, OCR

Used in: Hybrid RAG Portfolio Assistant

Frameworks: LangChain, Langfuse Observability, Custom Guardrails

Used in: Hybrid RAG Portfolio Assistant

Vector DBs: OpenSearch k-NN (HNSW, cosine similarity), Gemini Embedding-2 (768-dim)

Used in: Hybrid RAG Portfolio Assistant

Evaluation: RAGAS (Faithfulness, Relevancy, Precision, Recall)

Used in: Hybrid RAG Portfolio Assistant

Backend & API Development

Frameworks: FastAPI, Spring Boot (Java 17), Flask

Auth: JWT, OAuth 2.0, Spring Security

Used in: Smart Campus Operations Hub

Databases: PostgreSQL, MongoDB, Redis, MySQL

API Design: REST APIs, WebSocket (Real-time), Webhook Integration

Used in: Fraud Detection (WebSockets), RAG (Webhooks)

Frontend Development

Frameworks: React, Next.js 16, TypeScript, Tailwind CSS v4

Animation: Framer Motion

State: React Hooks, Context API

Programming & Development Tools

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Languages: Python, SQL, JavaScript/TypeScript, Java, Kotlin, R, PHP, Bash

Data: Pandas, NumPy, Spark, Hadoop, SQL, MongoDB

DevOps: Docker, Docker Compose, Kubernetes, AWS (EC2, S3, Lambda), Terraform

Version Control: Git, GitHub, GitLab, CI/CD Pipelines

Testing: Pytest, Evidently AI, Pandera

Cloud & Infrastructure

Cloud: AWS (Certified), GCP, Oracle Cloud Infrastructure

Containers: Docker, Docker Compose, Kubernetes, Coolify

Used in: RAG (6 containers), Fraud Detection (Docker Compose)

Orchestration: Apache Airflow, GitHub Actions

Used in: Fraud Detection (Automated MLOps)

Monitoring: Grafana, Prometheus, Langfuse, DAGsHub

Used in: Fraud Detection (Grafana), RAG (Langfuse)

Data Engineering & Analytics

ETL: Microsoft SSIS, Apache Airflow, Custom Pipelines

OLAP: SSIS, OLAP Cube, Power BI Dashboards

Used in: ETL & Data Warehousing Project

Feature Store: Feast (Online + Offline)

Used in: Real-Time Fraud Detection System

Message Queue: Apache Kafka

Used in: Real-Time Fraud Detection System

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Featured Projects - Applied Skills

Real-Time Fraud Detection System

Technologies: CatBoost, Feast, Kafka, Airflow, MLflow, Grafana, PostgreSQL, Redis, Docker

Production fraud detection pipeline processing 10K transactions/sec with sub-15ms latency. ROC-AUC 0.9938, PR-AUC 0.94. 3-tier decision framework (Auto-Block, Manual Review, Soft Audit). Online/offline feature store with Feast, automated drift detection with Evidently AI, weekly retraining via Airflow DAGs, real-time dashboard with WebSockets.

Hybrid RAG Portfolio Assistant

Technologies: OpenSearch, Gemini, Cohere, Docling, LangChain, FastAPI, Docker, Coolify

Production RAG system with hybrid search (BM25 + k-NN), parent-child chunking, Cohere rerank-v3.5 (Top 10 to Top 3), strategy router, input/output guardrails, portfolio-only classifier, conversation memory, LRU cache. Deployed on Oracle Cloud via Coolify with 6 containers. RAGAS evaluation.

Smart Campus Operations Hub

Technologies: Java 17, Spring Boot, Spring Security, MongoDB, React, OAuth 2.0

Full-stack campus operations with JWT + OAuth 2.0 authentication, facilities management, bookings, maintenance tickets, and notifications.

EcoSprout - Carbon Credit Marketplace

Technologies: Next.js, TypeScript, Tailwind CSS, Clerk, Convex, Turbo

Carbon credit marketplace connecting developers, verifiers, and buyers. Built educational hub with structured learning modules and community resources.

Movie Recommendation System

Technologies: Scikit-learn, Surprise, SBERT, Pandas, Jupyter

Comprehensive recommendation engine with KNN, SVD, SBERT content-based, and hybrid ensemble models on MovieLens data.